Who Waits Longer in Buffalo? Measuring Fairness in 311 Response Times Across Neighborhoods

Phase 1: Data Cleaning and Exploratory Data Analysis

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***Abstract*—Buffalo residents rely on the 311 system to report non-emergency issues such as potholes, missed trash pickups, and housing violations, but it is unclear whether every neighborhood receives the same level of service. This project examines whether 311 response times are equal across Buffalo's council districts and neighborhoods from 2020 to 2026. We merged the City of Buffalo's two 311 datasets into one dataset of over 560,000 requests, applied eight cleaning and processing operations that uncovered major data quality issues such as placeholder and swapped coordinates, and performed exploratory data analysis on request trends, request types, and resolution times. Our cleaned dataset provides a foundation for measuring whether wait times depend on the type of problem reported or where residents live.**

Keywords—311 service requests, data cleaning, exploratory data analysis, service equity, open government data

I. Introduction

Buffalo residents depend on 311 to report everyday non-emergency issues that you wouldn't call 911 for. Issues like potholes, missed trash pickups, sewer problems, and housing violations still need to be addressed, even though they aren't emergencies. But not every part of the city may get the same level of service, and some neighborhoods could be waiting much longer for their issues to be resolved. Using the City of Buffalo's 311 Service Request data from 2020 to 2026, our project looks at whether response times are equal across the city, or whether some communities are consistently left waiting.

The questions we want to answer are:

1) Which council districts and neighborhoods wait for the longest 311 service requests to be resolved?
2) Do response times differ by request type or department, and does that explain the gaps between areas?
3) Have these gaps gotten better or worse from 2020 to 2026?
4) When we account for how many residents live in each area, do some neighborhoods still stand out?

A. Background

311 service requests are the main way for managing non-emergency city services. In the City of Buffalo, the 311 system allows residents to report issues with public works, sanitation, and regulatory enforcement. While emergency services (911) handle immediate threats to public safety, the 311 system takes care of issues that affect how the city operates and impact a neighborhood's baseline quality of life, such as potholes, missed trash collections, localized sewer malfunctions, and structural housing code violations.

When a resident submits a request by phone, online, or through the BUFF311 app, the city creates an official record, and every request means spending real money, time, and labor to fix it. Because of this, 311 data can reveal whether service is distributed fairly. When cities have limited resources, financial problems, or unfair systems, some neighborhoods end up waiting much longer for basic city repairs than others. When some neighborhoods must wait a long time for city fixes, it causes major problems:

- **People stop trusting the city:** When residents report a problem and nothing happens, they lose faith in local government. When people stop calling, the city's data looks like there are no problems, even though the neighborhood is being ignored.
- **Small issues turn into expensive disasters:** Waiting too long to fix simple things makes them much worse and more expensive. A small pothole can grow big enough to pop car tires or destroy the whole street. Ignoring housing violations can cause buildings to rot, which lowers property values for the whole block.
- **It unfairly hurts vulnerable neighborhoods:** If the neighborhoods waiting for the longest are mostly low-income areas or minority communities, the city is making existing inequalities worse. It means families who are already struggling are forced to live with broken sidewalks, dirty streets, and unsafe housing for much longer than wealthier areas.

B. Contribution

We are looking at the City of Buffalo's official 311 service request records from 2020 to 2026 to see how fast the city fixes problems. We are comparing the wait times against city council districts, neighborhood lines, and how many people live in each area. Looking at these patterns, we want to find out if your wait time depends on what the problem is, or simply where you live. This will show us if the City of Buffalo shares its workers and money fairly, or if certain communities are always left waiting.

Our findings can help several groups. City leaders can use them to decide where to send crews and money, council members can use them to advocate for their districts, and residents can use them as proof when their neighborhood is

being overlooked. Our project also creates something that doesn't exist yet: one clean dataset that combines both of Buffalo's 311 systems from 2020 to 2026. While cleaning it, we found problems that would mislead anyone using the raw data, like over 309,000 requests stored with fake placeholder coordinates and about 10,000 with latitude and longitude swapped. This contribution is crucial because Buffalo has limited resources, and without clear data, service gaps stay invisible and keep hurting the same neighborhoods. Measuring these gaps is the first step toward fixing them.

II. Data Sources

This project uses live, externally hosted data loaded directly from the City of Buffalo's Open Data Portal, so the raw data is not included in our submission. Both datasets are published by the City of Buffalo and were accessed on September 26, 2026.

- **311 Service Requests (July 2008 to May 2024) [1]:** 1,129,274 rows and 34 columns. We filter this to requests opened on or after January 1, 2020, leaving 362,538 rows.
- **311 Service Requests (May 2024 to Present) [2]:** 202,398 rows and 28 columns, covering May 11, 2024, to present.

After merging, our combined dataset has 564,936 rows, well above the 2,000-row requirement. Because the data is live, row counts may be slightly higher if the notebook is run at a later date.

III. Data Cleaning and Processing

Our raw data needed a lot of cleaning before it could answer our questions. We performed eight distinct cleaning and processing operations, each documented with comments and markdown in our notebook. The operations run in the order below because some depend on earlier ones. We started with 1,331,672 total rows across both datasets and ended with a clean dataset of 523,933 rows and 17 columns.

A. Operation 1: Data Integration

What: Merged the City of Buffalo's two 311 datasets (July 2008 to May 2024, and May 2024 to present) into one dataset. *Why:* Our 2020 to 2026 time range is split across both files, and the two city systems store the same information under different column names. *How:* Filtered the older dataset to requests opened on or after January 1, 2020, renamed the newer dataset's columns to match the older one (e.g., Created Date to Open Date), kept only shared columns, added a Source column to track each row's original file, and stacked them with pd.concat(). *Result:* The older dataset went from 1,129,274 to 362,538 rows, and the merged dataset has 564,936 rows.

B. Operation 2: Removing Irrelevant Data

What: Removed test entries and columns that don't help answer our questions. *Why:* Test entries aren't real requests, and extra columns add clutter without adding value. *How:* Dropped 4 rows where the department was listed as "Test," and dropped repeated or unneeded columns like Location (same information as latitude and longitude), City and State (every row is Buffalo, NY), and Census Block (too specific for our analysis). *Result:* 564,932 rows, reduced from 24 to 18 columns.

C. Operation 3: Data Type Conversion

What: Converted columns to their correct data types. *Why:* Dates stored as text can't be used to calculate response times, which is the core of our project. *How:* Converted Open Date and Closed Date to datetime, Latitude and Longitude to numbers, and Duplicate Request to True/False. We also standardized Census Tract to one 6-digit format, since the old data wrote tracts as decimals (e.g., 16.02) and the new data used 6-digit codes (e.g., 001602). *Result:* 0 dates failed to convert, and 23,807 invalid tract codes were set to missing, leaving 126 valid census tracts.

D. Operation 4: String Cleaning

What: Cleaned up inconsistent text formatting and spelling in the Type and Subject (department) columns. *Why:* The same department appeared under different names, which would split its requests into separate groups. *How:* Removed the unnecessary "(Req_Serv)" suffix from request types, and combined department name variants (e.g., "Parking" and "Dept of Parking" both became "Parking Department," and "Mayor's Office" became "Office of the Mayor"). Missing departments were labeled "Unknown." *Result:* Each department now appears under one consistent name.

E. Operation 5: Fixing Swapped and Placeholder Coordinates

What: Corrected location errors in Latitude and Longitude. *Why:* Bad coordinates would put requests in the wrong place on maps and in location-based analysis. *How:* Swapped back 10,126 rows in the newer dataset where latitude and longitude were flipped. We also found 309,526 older requests (2020 to 2023) stored at exactly (43.0, -79.0), a fake default location, and set those to missing. Finally, we checked that all remaining coordinates fall within Buffalo's boundaries. *Result:* 0 coordinates outside Buffalo, and 235,413 rows with verified real coordinates. Rows without coordinates can still be analyzed by council district and neighborhood.

F. Operation 6: Handling Missing Data

What: Found and handled missing values column by column. *Why:* Both city systems wrote the text "UNKNOWN" instead of leaving cells blank, so pandas didn't recognize them as missing. *How:* Converted "UNKNOWN" and "UnAssigned" to true missing values. For the 160,152 rows missing a council district, we filled in the value from the Council District 2011 column, then dropped the 22,439 rows that still had no district, since comparing districts is the point of our project. Requests with no closed date are still open, so we kept them and flagged them with a new isOpen column. *Result:* 542,493 rows, with 14,332 open cases flagged. One limitation: 2011 district lines don't perfectly match current ones, so a small share of filled-in districts may be off.

G. Operation 7: Removing Duplicates

What: Removed repeat reports of the same issue. *Why:* Duplicates inflate request counts for some areas, which would skew comparisons between districts. *How:* We found 0 exact duplicate rows and 0 repeated case IDs, so we removed the 4,769 requests the city flagged as duplicates, plus 13,791 near-

duplicates (same request type, same address, same day), keeping the earliest report. This runs before Operation 8 so near-duplicates are matched on the original, specific request types. *Result:* 523,933 rows.

H. Operation 8: Category Standardization

What: Grouped similar request types into broader categories. *Why:* Many request types describe the same issue under different names (e.g., "Pot Hole," "Pothole Issue," and "Other Hole in Road"), which makes comparisons by type unreliable. *How:* Mapped each request type into one of 24 categories, like "Roads & Potholes" and "Snow Management," with anything unmatched labeled "Other." *Result:* Reduced Type from 308 unique values to 25 categories.

IV. Exploratory Data Analysis

We performed exploratory data analysis, following the approach described by NIST [3] and Tukey [4], on our cleaned dataset of 523,933 requests to understand when, where, and what residents report, and how long the city takes to resolve requests. Each operation below describes what we found and how we will use it in later phases. Graphs for every year are included in our notebook; representative figures are shown here.

A. Monthly Request Trends Across Years

What we did: Plotted the number of requests per month for each year from 2020 to 2026 on one line graph (Fig. 1). *What we found:* Requests follow a clear seasonal pattern, peaking in the summer months (June to August) in most years. The biggest outliers are January, November, and December of 2022, which line up with major winter storms that year. Note that 2026 only includes data through September. *How we'll use it:* We will add month and season as features, since time of year affects request volume and possibly response times. Weather events like those in 2022 may need to be flagged so they don't distort comparisons between years.

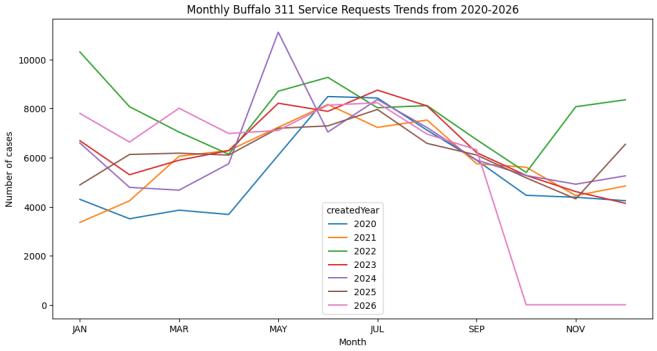


Fig. 1. Monthly Buffalo 311 service requests, 2020 to 2026. The 2026 line drops to zero after September because data for those months does not exist yet.

B. Monthly Trends Within Each Year

What we did: Plotted a separate monthly trend graph for each year; 2024 is shown in Fig. 2. *What we found:* Each year has its own spikes. For example, May 2024 has an unusually high 11,113 requests, which lines up with Buffalo's switch to its new 311 system that month. *How we'll use it:* We will compare response times before and after May 2024 to see whether the new system changed service, and treat the switch month carefully since it may reflect system changes rather than real issues.

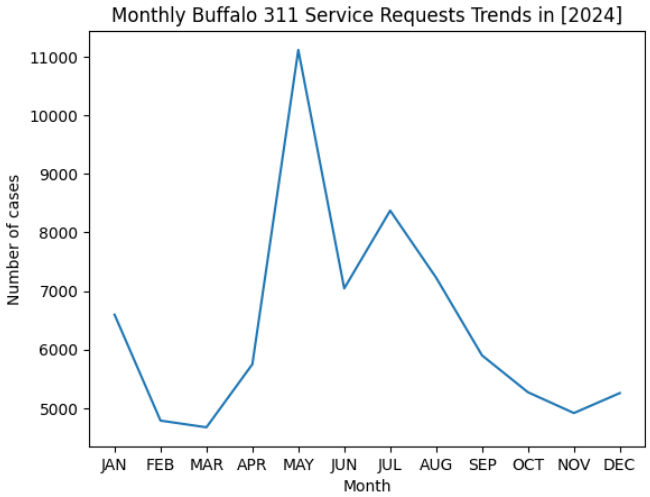


Fig. 2. Monthly Buffalo 311 service requests in 2024.

C. Top 5 Request Types Per Year

What we did: Plotted the five most common request categories for each year as bar charts; 2022 is shown in Fig. 3. *What we found:* Regular & Bulk Collection, Totes & Bins, and Violations & Inspections are in the top three almost every year. Snow Management entered the top five only in 2022, matching the storms found in Section IV-A. *How we'll use it:* Because request types dominate differently by year, we will compare response times within the same category, not just overall, so one common request type doesn't skew district comparisons.

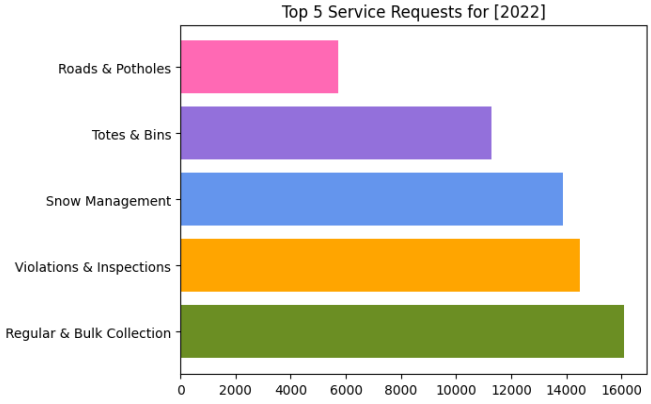


Fig. 3. Top five 311 request categories in 2022.

D. Total Requests Per Council District

What we did: Counted and plotted total requests for each of Buffalo's nine council districts (Fig. 4). *What we found:* Lovejoy has the most requests (77,491), followed by Fillmore (67,339) and Masten (64,728), while North has the fewest (48,636). *How we'll use it:* Raw counts don't account for population, so we will join Census population data by tract to

compare requests per resident, which is fairer across districts of different sizes.

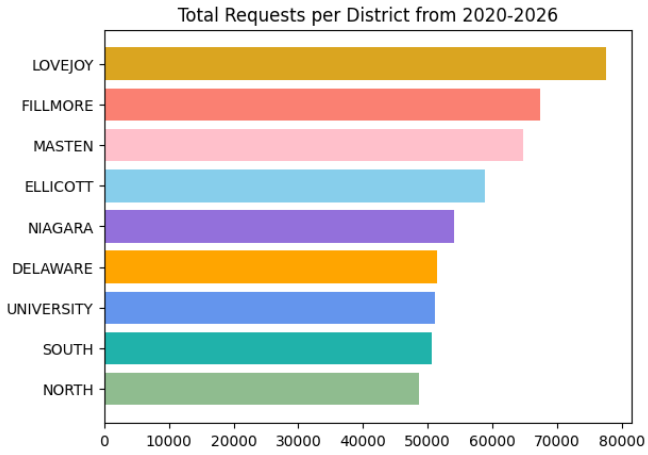


Fig. 4. Total 311 requests per council district, 2020 to 2026.

E. Resolution Times by Department

What we did: Created a new feature, resolution time in days (Closed Date minus Open Date), and summarized the mean and median for each department (Table I). *What we found:* Resolution times are heavily right-skewed: the overall median is about 4 days, but the mean is about 15 days, with some requests taking over 1,400 days. National Grid has the longest median (20.1 days), while the Parking Department resolves most requests in under a day (0.7 days). Public Works, Parks, & Streets handles the largest share of requests, with a median of 3.8 days. *How we'll use it:* Because of the skew, we will use the median instead of the mean when comparing areas and consider a log transform for modeling. Since departments differ so much, we will control for department when comparing districts.

TABLE I. RESOLUTION TIMES BY DEPARTMENT (DAYS)

Department	Mean	Median	Cases
National Grid	22.4	20.1	5,455
City Clerk	71.4	15.0	499
Buffalo Sewer Authority	20.6	13.2	3,729
Office of the Mayor	62.7	12.9	2,210
Buffalo Water	19.6	12.9	3,854
Buffalo Municipal Housing Authority	15.7	10.9	1,906
Utilities	15.5	10.7	15,570
Buffalo Fire Department	12.9	10.6	680
Permit & Inspection Services	22.1	8.0	77,172
Office of Strategic Planning	58.8	7.9	885
Public Works, Parks, & Streets Department	13.8	3.8	342,026
Assessment & Taxation	10.5	3.0	735
Buffalo Police Department	11.5	1.8	24,630
Law Department	10.8	1.1	390
Parking Department	2.6	0.7	30,457

F. Top 10 Census Tracts by Request Volume

What we did: Ranked Census tracts by total requests (Table II). *What we found:* Tract 000200 has the most requested (15,030), and each of the top 10 tracts has over 8,500 requests. *How we'll use it:* We will check whether high-volume tracts are also the slowest to be served and use tracts to link our data to Census demographics like population and income.

TABLE II. TOP 10 CENSUS TRACTS BY REQUEST VOLUME

Census Tract	Requests
000200	15,030
002400	12,729
004300	11,210
003700	10,721
004500	10,481
001000	10,213
004100	10,187
006100	10,026
005500	8,530
004200	8,517

G. Top 10 Addresses by Request Volume

What we did: Combined address number and street into a full address and ranked the most frequent ones (Table III). *What we found:* The top address is 65 Niagara Sq (1,694 requests), which is Buffalo City Hall. Many others are labeled "INTERSECTION" with only a street name. These are likely default or placeholder locations rather than actual problem sites. *How we'll use it:* We will flag or exclude these placeholder addresses from any location-based analysis, since they would falsely make some areas look like hotspots.

TABLE III. TOP 10 ADDRESSES BY REQUEST VOLUME

Address	Requests
65 NIAGARA SQ	1,694
INTERSECTION Main St	849
INTERSECTION Hertel Ave	629
INTERSECTION Bailey Ave	480
INTERSECTION Broadway St	457
414 PARKSIDE	456
275 SUFFOLK	446
INTERSECTION Genesee St	443
INTERSECTION Clinton St	435
INTERSECTION Amherst St	431

H. Request Types by Department

What we did: Built a heatmap showing how the top 10 request types are spread across the five busiest departments (Fig. 5). *What we found:* Public Works, Parks, & Streets handles the large majority of requests (351,965 requests), especially collection, totes, and road issues, while Permit & Inspection Services mainly handles violations. *How we'll use it:* Since request type and department are closely linked, we will

be careful not to include both as separate features without accounting for their overlap.

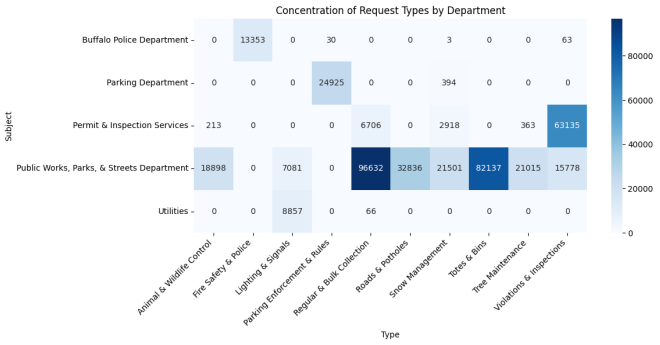


Fig. 5. Concentration of request types by department.

V. Bonus Section: Novelty of Our Project

Research on 311 systems has grown, but it leaves open questions our project is set up to answer. Early studies focused mostly on who reports problems, not on how fast cities fix them. For example, a study of Boston's 311 requests examined disparities by race, education, and income in making service requests, and found that poorer neighborhoods were less likely to use 311, except for smartphone reporting [5]. Similarly, Minkoff [6] studied Census-tract-level differences in 311 contact volume across New York City.

More recent work shows why request volume alone can be misleading. In Kansas City, researchers found that despite greater objective and subjective need, low-income and minority neighborhoods were less likely to report street condition or nuisance issues [7]. This means an area with fewer requests isn't necessarily an area with fewer problems, which is why our project focuses on response times and keeps Census tract data so we can compare areas fairly by population.

The closest study to our examined response times directly. Using about 42 million 311 requests from 13 large cities between 2011 and 2019, Hamel and Holliday [8] found no evidence that cities respond faster to the same type of request from whiter and more affluent neighborhoods. However, on average, white and wealthy neighborhoods still received faster responses, and the authors suggest this may be because non-white and poor neighborhoods tend to request services that take more time and resources. This finding directly shapes our second research question. Our project builds on this research in four new ways:

1) **A mid-sized city:** Most existing studies focus on large cities like New York, Boston, and Kansas City. Buffalo is a smaller city with its own budget limits and neighborhood history, and to our knowledge its 311 response times have not been studied this way.
2) **A before-and-after comparison:** Buffalo switched 311 systems in May 2024. Because our dataset spans both systems from 2020 to 2026, we can check whether the new system changed response times or the gaps between neighborhoods.
3) **A combined, cleaned dataset:** Buffalo's 311 data is split across two systems with different formats. We merged them into one dataset and documented hidden data problems, including over 309,000 requests stored at a fake placeholder location and about 10,000 with latitude and longitude swapped, which would mislead any analysis of the raw data.
4) **Testing the request-type explanation locally:** Following Hamel and Holliday [8], we compare response times both overall and within the same request category, so we can tell whether any gaps in Buffalo come from what residents request or where they live.

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